4.1.2. PPAR Agonists: Pioglitazone and Fenofibrate

Pioglitazone, a thiazolidinedione, is a potent ligand of the nuclear receptor PPAR-γ (peroxisome proliferator-activated receptor-gamma). Although primarily employed as an antidiabetic agent, it may also possess anti-inflammatory properties. Mechanistically, PPAR-γ activation attenuates the expression of pro-inflammatory genes, thereby reducing the production of inflammatory cytokines, matrix metalloproteinases (MMPs), and reactive oxygen species (ROS) [Shahin2011]. A further rationale is the bidirectional relationship between insulin resistance and inflammation, whereby each process reinforces the other [Ormseth2013].

In the adjuvant-induced arthritis (AIA) rat model, oral pioglitazone reduced disease severity, but only at the top dose of 30 mg/kg/day [Koufany2008]. It reduced bone erosions and inflammation-associated bone loss but failed to prevent cartilage lesions [Koufany2008]. Importantly, these anti-arthritic doses exceeded those needed to restore insulin sensitivity and were accompanied by increased adiposity [Koufany2008]. In a later study using the same model, pioglitazone monotherapy (1.35 mg/kg/day) had no significant effect on disease activity, whereas its combination with methotrexate (MTX; 0.225 mg/kg/day) suppressed disease activity more than either agent alone ($P < 0.05$) [Roy2017].

At the clinical level, an Egyptian non-randomized, single-blind trial enrolled 49 patients with rheumatoid arthritis (RA) and type 2 diabetes mellitus, all maintained on MTX 15 mg/week [Shahin2011]. The 28 patients with more active disease added oral pioglitazone 30 mg/day for 12 weeks, while the other 21 took placebo [Shahin2011]. Compared with placebo, pioglitazone produced significantly larger reductions in the DAS28 disease activity score ($P = 0.001$) and in C-reactive protein (CRP; $P < 0.0001$), although weight gain was reported in 25% of treated patients [Shahin2011].

Further evidence came from a US crossover trial that was randomized, double-blind and placebo-controlled, in which 34 patients with RA on stable therapy received pioglitazone 45 mg/day and placebo for 8 weeks each [Ormseth2013]. Pioglitazone produced a modest 9.3% reduction in CRP-based DAS28 ($P = 0.046$) but no improvement in DAS28 based on the erythrocyte sedimentation rate (ESR) ($P = 0.92$) [Ormseth2013]. Lower-extremity edema occurred in 16% of participants while on pioglitazone but in none while on placebo [Ormseth2013].

A larger crossover trial with the same design and dose enrolled 143 non-diabetic patients with RA, with 3 months per treatment arm [Marder2013]. DAS28-CRP decreased by a mean of 0.31 on pioglitazone compared with 0.15 on placebo (P = 0.02) [Marder2013]. Nevertheless, the authors concluded that these findings did not, at that stage, support adding PPAR-γ agonists to the standard treatment of RA [Marder2013]. Overall, the observed therapeutic effect was modest and appeared to be driven primarily by improvements in CRP-based measures rather than by consistent reductions in joint counts. When weighed against the risks of fluid retention and weight gain, these findings do not support the routine use of pioglitazone in RA; however, it may warrant further investigation as a selective adjunctive therapy for patients with coexisting insulin resistance or type 2 diabetes mellitus.

Fenofibrate is a widely used PPAR-α agonist prescribed primarily for the treatment of hypertriglyceridemia [Zhou2012]. In vitro, fenofibrate inhibited Th17-cell differentiation in murine T cells in a concentration-dependent manner over the range of 5–20 μM, producing a 74% reduction at 20 μM [Zhou2012]. However, other PPAR-α agonists, including WY14643, GW7647, and bezafibrate, had no effect on Th17-cell differentiation, suggesting that the inhibitory effect of fenofibrate may be independent of PPAR-α activation [Zhou2012]. In rats with adjuvant-induced arthritis, 12 days of fenofibrate treatment (300 mg/kg/day) reduced the arthritis score from 11.0 to 5.83, corresponding to an approximately 47% reduction in the external signs of inflammation (P < 0.01) [Castillero2012]. In patients with RA, the only clinical data come from one randomized trial in which fenofibrate performed similarly to a statin on ESR, CRP, and DAS28 [Xing2015]. Placebo-controlled trials are therefore needed to determine whether these preclinical anti-inflammatory effects translate into meaningful therapeutic benefits in RA.

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